

EDIDIONG DAVID IBOKETE

DATA SCIENTIST | MACHINE LEARNING RESEARCHER | APPLIED AI ENGINEER

Houston, TX | iboketee@icloud.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data scientist, machine learning researcher, and applied AI engineer with about three years of experience building reliable data, ML, and AI systems across IoT cybersecurity, retrieval-augmented generation, forecasting, institutional analytics, and geospatial operations. Co-author of two submitted security-ML research papers and builder of a deployed RAG assistant with embedding retrieval, 17 evaluation suites, 277 audit questions, feedback-driven testing, FastAPI, Azure App Service, and GitHub Actions-backed deployment. Combines Python, SQL, statistical modeling, data engineering, cloud deployment, and rigorous evaluation to turn complex requirements into reproducible decision-support systems.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, SAS, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, XGBoost

Machine Learning & Statistics: Regression, classification, time-series forecasting, anomaly detection, clustering, feature engineering, threshold calibration, SHAP, hypothesis testing, model validation, explainability

Applied AI & Evaluation: RAG, embeddings, semantic retrieval, LLMs, prompt engineering, knowledge-base design, retrieval benchmarks, top-k evaluation, feedback loops

Engineering & MLOps: FastAPI, Azure App Service, REST APIs, Git/GitHub, CI/CD (GitHub Actions / GitHub Pages), YAML/Markdown, configuration-driven experiments, automated notebook execution, reproducible pipelines

Data Engineering & BI: ETL, Alteryx, MS SQL Server, MySQL, SLATE CRM, data validation, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Center for Research Libraries | Chicago, IL

Algorithm & Data Engineer Intern

Jun 2025 - Aug 2025

- Translated ambiguous cooperative-cataloging requirements across 19 institutions into a Python/Pandas capacity-constrained matching pipeline covering 194 tasks and scoring priority, qualification fit, reciprocity, trust, and contributor capacity.
- Engineered normalized requester and contributor profiles from inconsistent survey inputs, producing reviewable scoring, assignment, and diagnostic outputs for stakeholder evaluation.
- Built an anonymized reproducibility package that generated 565 candidate matches, passed end-to-end smoke testing, and converted evolving workflows into documented allocation rules and validation checkpoints.

Canisius University | Buffalo, NY

Graduate Assistant, Data Analytics & Reporting

Aug 2024 - Dec 2025

- Engineered automated ETL and validation workflows with SQL, SAS PROC SQL, R, and Alteryx across SLATE CRM and MS SQL Server for enrollment reporting and institutional analytics.
- Removed duplicate and inconsistent registration records with 100% deduplication accuracy; optimized relational queries and validation logic to reduce record redundancy by 70%.
- Developed Tableau and Power BI dashboards and documented data flows, reporting logic, and validation checkpoints for planning, maintainability, and stakeholder communication.

Geospatial Intelligence and Data Analysis Centre, Ministry of Defense | Abuja, Nigeria

Pioneer Database Administrator & Geospatial Data Analyst

2021 - 2023

- Administered MySQL and MS SQL Server environments and secure access controls for structured and geospatial datasets, maintaining integrity and reporting availability in mission-critical operations.
- Improved query efficiency by 30% through indexing and query restructuring under high-demand workloads.
- Built Python-assisted analytical workflows that accelerated monthly reporting by 25% and integrated geospatial and structured data into analytics-ready decision-support pipelines.

SELECTED RESEARCH & APPLIED AI PROJECTS

Certified Anomaly Detection for IoT Security | Co-Author, Research Software & Evaluation

- Implemented and evaluated a Negative-Binomial concentration framework for overdispersed IoT traffic, achieving 1.000 AUC, 0.000 certified-mode benign FPR, and 448 microseconds latency per 10 ms window; NB fit better than Poisson on 98.5% of applicable Gotham devices.
- Developed the Poisson-CI evaluation pipeline, attaining 0.9648 ROC-AUC and 0.9708 F1 at a matched 1% FPR across eight attack families, with classical baselines and cross-dataset validation.
- Owned research-lifecycle components across feature engineering, model integration, benchmarks, matched-FPR evaluation, ROC and latency analysis, result verification, publication artifacts, configuration-driven experiments, automated notebook execution, and reproducibility packaging.

Project ARIEL - Deployed RAG Portfolio Assistant | Azure App Service | FastAPI | OpenAI

- Designed and deployed an end-to-end RAG application connecting a curated YAML/Markdown knowledge base, OpenAI embeddings, semantic retrieval, grounded generation, a FastAPI backend, and a JavaScript widget embedded across a GitHub Pages portfolio.
- Built and maintained 17 retrieval evaluation suites covering 277 audit questions; the latest audit achieved 100% top-1 and 100% top-k retrieval accuracy.
- Implemented feedback-driven quality controls using targeted patch tests, smoke tests, similarity-score review, automated reports, and failure analysis; deployed through Azure App Service and GitHub Pages build/deployment workflows.

Commercial Building Energy Forecasting & Optimization | Predictive Modeling Lead | 1st Place

- Built an XGBoost forecasting pipeline with time-series, building, and weather features, achieving 0.9804 R-squared, 5.44 MAE, and 34.28 RMSE on future-period evaluation across a 1,600+ building commercial portfolio.
- Used SHAP, residual diagnostics, and savings simulations to identify 10 underperforming buildings and estimate \$858K-\$1.7M in annual savings from 10%-20% portfolio-wide energy reductions.

Additional Machine Learning & Analytics

- Modeled purchase propensity across 12,205 cleaned sessions with Gradient Boosting, achieving 0.9377 ROC-AUC and 75.13% recall at a tuned 0.70 threshold; compared classification models on breast-cancer data and achieved 98.25% test accuracy and 95.24% malignant-case recall.
- Built a TensorFlow/Keras housing regressor with 0.961 R-squared and approximately \$11.6K MAE, then implemented K-means from scratch and Gaussian anomaly detection with F1-optimized threshold selection.
- Published three interactive Tableau dashboards spanning airline operations, workforce analytics, and location intelligence using calculated fields, filters, parameters, tooltips, and dashboard-level KPI design.

EDUCATION & CERTIFICATIONS

Master of Science, Data Analytics (Big Data)

Completed Dec 2025

Canisius University, Buffalo, NY

Bachelor of Science, Economics | University of Abuja, Nigeria

Microsoft Certifications: Machine Learning Operations Engineer Associate | Azure Fundamentals | Power BI Data Analyst Associate

RESEARCH COMMUNICATION & RECOGNITION

Research Papers: "Certified Anomaly Detection for Overdispersed IoT Traffic via a Negative-Binomial Khinchine-Fubini Concentration Inequality" (submitted under review, 2026) | "Certified Real-Time Anomaly Detection for IoT Networks with Pre-Deployment Verification" (submitted, 2026)

Presentations: "From Probability to Protection: Poisson Inequalities for Machine Learning in Cybersecurity" (2025) | "Using AI to Address Cataloging Backlogs" (2025)

Recognition: 1st Place, Data Masters Challenge (2025) | Koessler Summer Fellowship (2025) | Dean's List Excellence Award